SECTION A – Answer all questions. Circle the correct option on the question sheet and return it.

1. Which of the following is not a valid variable name declaration?
   A. int _a3; B. int a_3;
   C. int _3-a; D. int _3a

2. Which of the following is the correct order of evaluation of $z = x + y * z/4\%2 - 1$?
   A. * / % + - = B. = * / % + -
   C. / * % - + = D. * % / - + =

3. Which operator has the lowest priority?
   A. ++ B. %
   C. + D. ||

4. What will be the output of the following program?

```
...
    double x = 28;
    int r;
    R = x%5;
    cout<<"R ="<<r;
...
```

   A. R = 3
   B. Run time Error
   C. Compile time Error
   D. None of the Above

5. Which of the following is not a keyword in C++?
   A. continue B. break
   C. int D. function

6. In which of the following does the loop body executes at most once under test condition that runs to false?
   A. While loop construct B. For loop construct
   C. Do While loop construct D. B and C

7. What will be the output of the following program?

```
...
  int x=10,y=20;
  cout<<x<<"\n"<<y;
...
```

   A. 10
   B. 20
   C. 10
      20
   D. None of the above

8. _____ is a step-by-step procedure of solving a problem
   A. Algorithm B. Random select process
   C. Monte Carlo process D. None of the above

9. What will be displayed on the screen when the C code below is run?

```
...
    cout<<"Hello Ghana\!";
    return (0);
...
```

   A. Hello Ghana
   B. Hello Ghana!
   C. hello Ghana!
   D. None of the above

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10. Which pre-processor directive will you add in the following program to make it work correctly?

```
_______________ //pre-processor
...
cout<<" Value is:\n"<<"\t"<< log(36.0));
...
```

A. #include <conio.h>
B. #include <cmath>
C. #include <cstring>
D. #include <clog>

Section B – Answer all questions

(1) *Describe the following*

I. Interpreter

_____

_____

_____

_____

_____

II. Assembler

_____

_____

_____

_____

_____

_____

III. Compiler

_____

_____

_____

_____

_____

_____

IV. Define software engineering

_____

_____

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V. List and explain two types of programming error you know.

_____

_____

_____

_____

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(2) *Fill in the blanks in each of the following:*

a)_____ are used to document a program and improve its readability.

b) The object used to print information on the screen is _____.

c) A C++ statement that makes a decision is _____.

d) Calculations are normally performed by _____ statements.

e) The _____ object inputs values from the keyboard.

a) What arithmetic operations are on the same level of precedence as multiplication? _____.

b) When parentheses are nested, which set of parentheses is evaluated first in an arithmetic expression?

_____.

c) A location in the computer's memory that may contain different values at various times throughout the execution of a program is called a _____.

(3) *State which of the following are true and which are false. If false, explain your answers.*

a) C++ operators are evaluated from left to right.

_____

_____

_____

b) The following are all valid variable names: `_under_bar_`, `m928134`, `t5`, `j7`, `her sales`, `his_account_Total`, `a`, `b`, `c`, `z`, `z2`.

_____

c) The statement `cout << "a = 5;";` is a typical example of an assignment statement.

_____

d) A valid C++ arithmetic expression with no parentheses is evaluated from left to right.

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(4) ***Select the correct options***

I. Which of the following C++ statements contain variables whose values are replaced?

a) `cin >> b >> c >> d >> e >> f;`

d) `cout << "a = 5";`

II. Given the algebraic equation $y = ax^3 + 7$, which of the following, if any, are correct C++ statements for this equation?

a) y = a * x * x * x + 7;
b)
c)
d)
e) y = a * (x * x * x) + 7;
f) y = a
g)

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III.

a) _____

b) _____

// _____

Section C: ***Answer only one question in the section***

Question 1

Write a C++ program that allows the integers 1, 2, 3,… to be added until the sum exceeds 20 million. Your program should also display the last number that was added to the sum.

Question 2

Write a C++ program that reads in two integers and determines and prints if the first is a multiple of the second.

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